

Naman Gupta

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EDUCATION

University of Southern California

MS in Computer Science

Los Angeles, USA

Jan 2026 – Dec 2027

Graphic Era Deemed to be University

B.Tech in Computer Science and Engineering

Dehradun, India

Sep 2021 – June 2025

INTERNSHIP/WORK EXPERIENCE

Morgan Stanley

Mumbai, India

Technology Analyst

Aug 2025 – Nov 2025

- Engineered a backend performance overhaul by re-architecting query paths and caching layers across critical internal applications, reducing average response time by **30%** and improving system uptime.
- Developed a suite of reusable UI components for real-time monitoring dashboards by shipping internal tooling across engineering and operations teams, eliminating **15+** hours of manual reporting work per week for stakeholders.

Technology Apprentice

July 2024 – Aug 2025

- Designed a data-driven loan routing engine by building a recommendation model that surfaces the top 3 best-fit employees per case, cutting average task completion time by **20%** across loan processing teams.
- Built predictive workload distribution models trained on **12** months of throughput and staffing data, raising balanced resource allocation accuracy across teams by **25%** while reducing employee burnout risk.

RESEARCH WORK

Deep Learning Approach for Malicious URL Detection | IEEE Paper

Feb. 2025

- Engineered a malicious URL detection system by training and benchmarking four deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) on a dataset of over **450,000** labeled URLs, hitting **99.25%** precision with Bi-LSTM.
- Conducted a comparative architecture analysis by benchmarking bidirectional against unidirectional context for sequential URL pattern recognition, improving classification F1-score by **4.8%** for more robust phishing detection.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, Python, SQL, HTML, CSS

Developer Tools: VS Code, Git, GitHub, Docker, Postman, Linux, Microsoft, PyCharm, Jupyter Notebook

Framework / Library: Node.js, ReactJS, FastAPI, Flask, Django, TensorFlow, Keras, Pandas, OpenCV

Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Natural Language Processing, Agentic AI

PROJECTS

TechHire: AI Job Search Platform [on GitHub](#) | Python, FastAPI, React, PostgreSQL

May 2026 – June 2026

- Built a full-stack tech job search platform by integrating a Python scraper with a FastAPI backend and React frontend, indexing over **5,000** live postings with sub-second search across skills and salary filters.
- Integrated a multi-model Groq LLM pipeline into an AI resume analyzer that synthesizes job summaries across three models, substantially improving candidate-job fit scoring and resume quality feedback.

DNA Sequence Alignment: DP vs. Divide-and-Conquer [on GitHub](#) | Python, Algorithms

Apr 2026 – May 2026

- Implemented a novel DNA sequence alignment solution using the Needleman-Wunsch algorithm, building a full **$O(m \times n)$** dynamic programming table with configurable gap penalties and a custom mismatch cost matrix.
- Optimized memory usage by engineering Hirschberg's divide-and-conquer algorithm, cutting peak memory footprint by **60%** from **$O(m \times n)$** down to **$O(m + n)$** without sacrificing alignment accuracy or traceback correctness.

Real-Time Sign Language Detection [on GitHub](#) | Python, OpenCV, TensorFlow, Keras

Feb 2026 – March 2026

- Developed a real-time computer vision pipeline leveraging OpenCV for gesture capture, processing live hand gesture video streams at **30** frames per second with minimal detection latency and jitter.
- Trained a TensorFlow/Keras CNN by fine-tuning across **26** gesture classes, achieving **98%** real-time translation accuracy for practical, everyday sign language communication and use.

Mind Map: Mental Wellbeing Detection [on GitHub](#) | Python, BERT, LLaMA, Web Scraping

Dec 2024 – June 2025

- Created a mental health screening pipeline by web-scraping over **50,000** Reddit posts and comments, structuring the data for downstream sentiment analysis across mental-health-focused online communities.
- Fine-tuned BERT and LLaMA models on labeled sentiment data, achieving **91%** classification accuracy for real-time wellbeing screening across thousands of scraped community posts daily.